EXERCISES

Exercise sheet 5

AI504 — Knowledge Representation

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Contents

1 Exercise 1^+

Direct methods are able to use only zero-order information, that is, only evaluations of f. How many evaluations are needed to approximate the derivative and the Hessian of an n-dimensional objective function using finite difference methods? Why do you think it is important to have zero-order methods?

1.1 Solution

In higher dimensions it is very computationally expensive to compute ∇f and `nnf`. Because of this we resolve to zero-order methods (like methods that use distribution)

2 Exercise 2

2.1 Solution

3 Exercise 3^*

The Nelder-Mead algorithm has three parameters α, β, and γ. How would you approach the problem of tuning these parameters?

3.1 Solution

- α

  scaling factor for the reflection step

  $$x_r = x_m + \alpha \cdot x_m - x_h$$

- β

  scaling factor for the expansion step

  $$x_e = x_m + \beta \cdot (x_r - x_m)$$

- γ

  scaling factor for the contraction step

  $$x_c = x_m + \beta \cdot (x_h - x_m)$$

4 Exercise 4^*

Consider the natural evolutionary strategy for an univariate function. Assume the univariate normal distribution as proposal distribution $p(x|\theta) = \mathcal{N}(x|\mu, \sigma^2)$

- Derive the update rule for θ
- after a number of iterations the value of μ becomes equal to x_*, that is, the minimum of f, what will be the update rule for σ^2 and what will be the difficulty encountered by the algorithm?

4.1 Solution

We iteratively search for a good distribution θ

$$\theta_{k+1} = \theta_k + \alpha \nabla_\theta \mathbb{E}_{x \sim p(\cdot|\theta)}[f(x)]$$

where

$$\mathbb{E}_{x \sim p(\cdot|\theta)}[f(x)] = \int f(x) p(x|\mu, \sigma^2) \ \mathsf{dx} \approx \frac{1}{m} \sum_{i=1}^{n} \frac{f}{x_i}$$

Idearly, we look for

$$(\mu, \sigma^2) \to (x^*, 0)$$

So lets rewrite this

$$\nabla_\theta \mathbb{E}_{x \sim p(\cdot|\theta)}[f(x)] = \int f(x) p(x|\mu, \sigma^2) \ \mathsf{dx}$$

$$\text{since } \nabla_\theta \log(p(x|\theta)) = \frac{1}{p(x|\theta)} \cdot \nabla_\theta p(x|\theta)$$

$$\int f(x) p(x|\mu, \sigma^2) \ \mathsf{dx} = \int f(x) \nabla_\theta \log(p(x|\theta)) p(x|\theta)$$

$$\text{using the same rule as before for expected values,}$$

$$\mathbb{E}_{x \sim p(\cdot|\theta)}[f(x) \nabla_\theta \log(p(x|\theta))] \approx \frac{1}{m} \sum_{i=1}^{n} f(x_i) \nabla_\theta \log(p(x_i|\theta))$$

5 Exercise 5^*

5.1 Solution

6 Exercise 6^*

6.1 Solution